

Lab 4 - Wazuh SIEM / Log Analysis

Windows Authentication Event Collection and Investigation

Attribute	Details
SIEM platform	Wazuh
Server	Ubuntu Server 24.04 LTS
Endpoint	Windows 11 Pro
Architecture	Wazuh Manager + Indexer + Dashboard + Windows Agent
Network	VMware NAT

1. Objective

Deploy a small Wazuh SIEM environment, connect a Windows 11 endpoint using the Wazuh agent, generate controlled authentication activity, and investigate Windows security events in Threat Hunting.

2. Architecture

Endpoint activity -> Windows Event Logs -> Wazuh Agent -> Wazuh Manager / Indexer -> Wazuh Dashboard -> Analyst investigation.

3. Deployment and Enrollment

- Installed the Wazuh all-in-one stack on Ubuntu Server.
- Enrolled a Windows 11 endpoint and verified the agent as active.
- Confirmed events were searchable in the Wazuh dashboard.

4. Failed Logon Investigation - Event ID 4625

Field	Observed interpretation
Event ID	4625 - failed Windows logon
Logon Type	2 - interactive/local logon
Source IP	127.0.0.1 - localhost
Authentication package	Negotiate
Rule level	5 - moderate Wazuh severity

The evidence was consistent with the controlled lock-screen authentication test and did not indicate a remote brute-force source. Automated ATT&CK tags were treated as product metadata rather than proof of adversary behavior.

5. Successful Logon Investigation - Event ID 4624

Field	Observed interpretation
Event ID	4624 - successful logon
Logon Type	7 - workstation unlock
Target domain	MicrosoftAccount
Process	lsass.exe
Rule level	3 - low-severity Wazuh alert

6. Event Comparison

Event	Result	Meaning
4625	Failure	Interactive/local sign-in failed
4624	Success	Workstation unlock succeeded

7. Analyst Assessment

Several failed local authentication attempts were followed by a successful workstation unlock. No evidence of remote access, malware execution, or account compromise was identified from the reviewed events.

8. Skills Demonstrated

- Wazuh deployment
- Windows agent enrollment
- SIEM ingestion
- Threat hunting
- Windows authentication analysis
- Agent troubleshooting
- Evidence-based interpretation

9. Conclusion

The lab demonstrated an end-to-end SIEM workflow from endpoint activity to centralized alert investigation while emphasizing the need to validate automated classifications against the underlying telemetry.